Recommendation System for Olist E-commerce Retailer

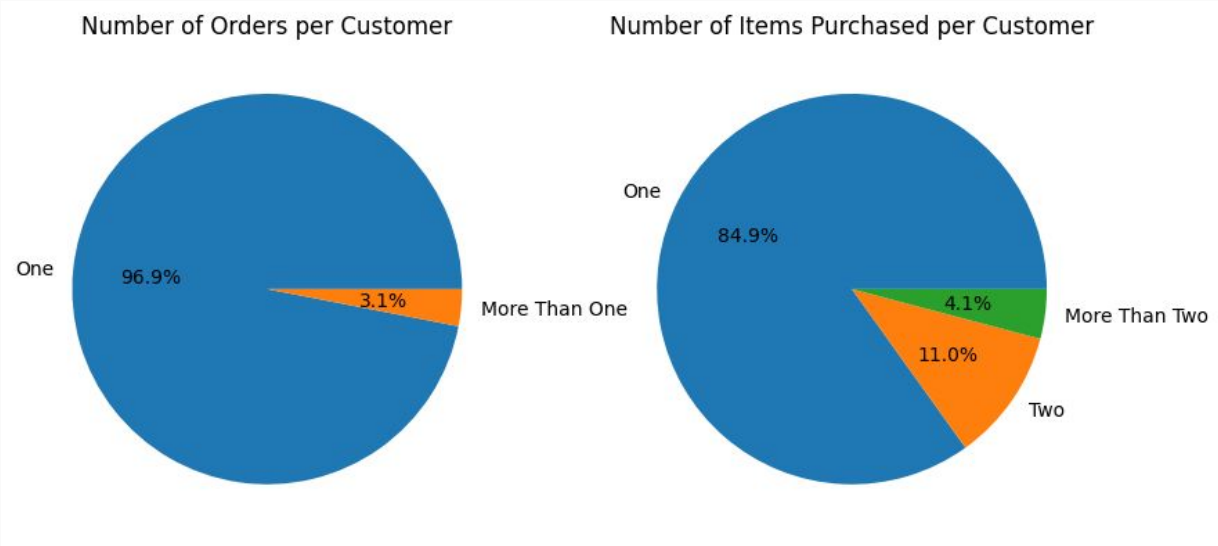
NYU x Peak Datathon

Blow Up the Pipeline **Team**

Purchase History

– Most customers only ordered once & purchased one item

Very limited amount of validation data prompt us to adopt simple approaches over complex time-series based models.



Framework

The customer-product interactions are **sparse** in the Olist dataset, with 85% customers only having one purchase record.

Therefore we construct our recommendation system from three aspects:

1. Customer-to-customer recommendation based on Bayesian Classifiers
2. Product-to-product recommendation based on Category Similarity
3. Intrinsic product features

Geographical Information Matters

— A geovisualization of two representative sellers and their customers

Seller in the **south** & its customers



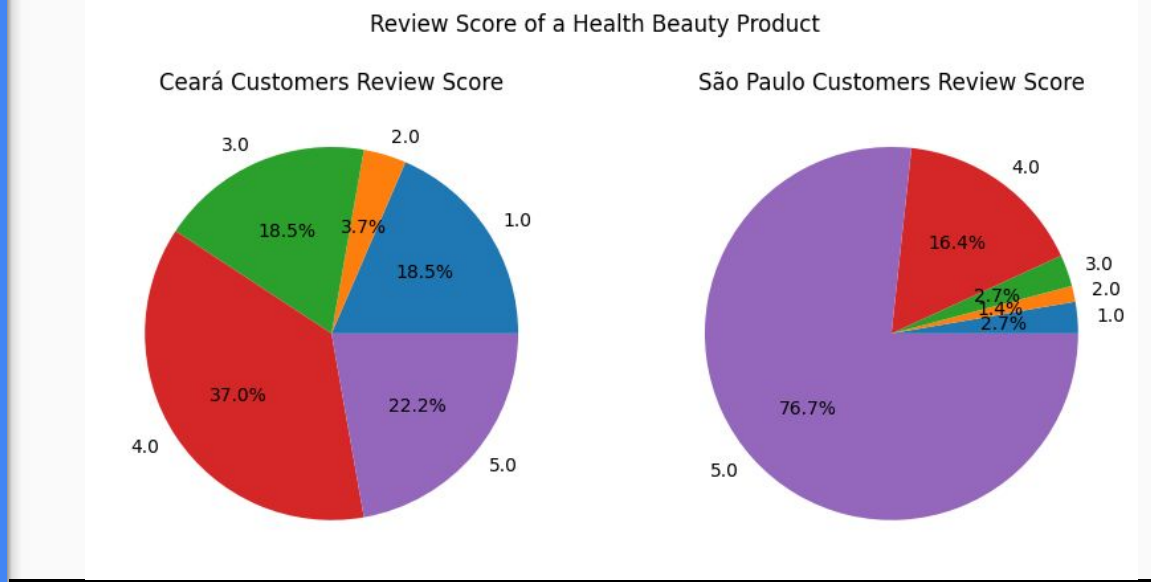
Seller in the **north** & its customers

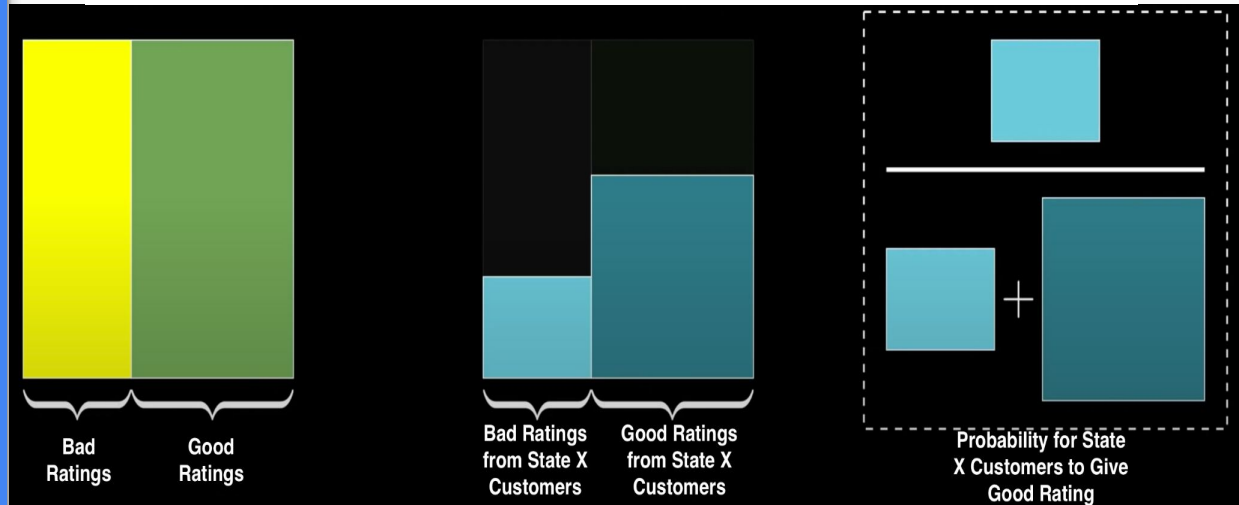


Part 1: Bayesian Inference

Recommend products that are either:

- Popular in general, or
- Especially favored by customers from the same state.





Part 2: Category Similarity

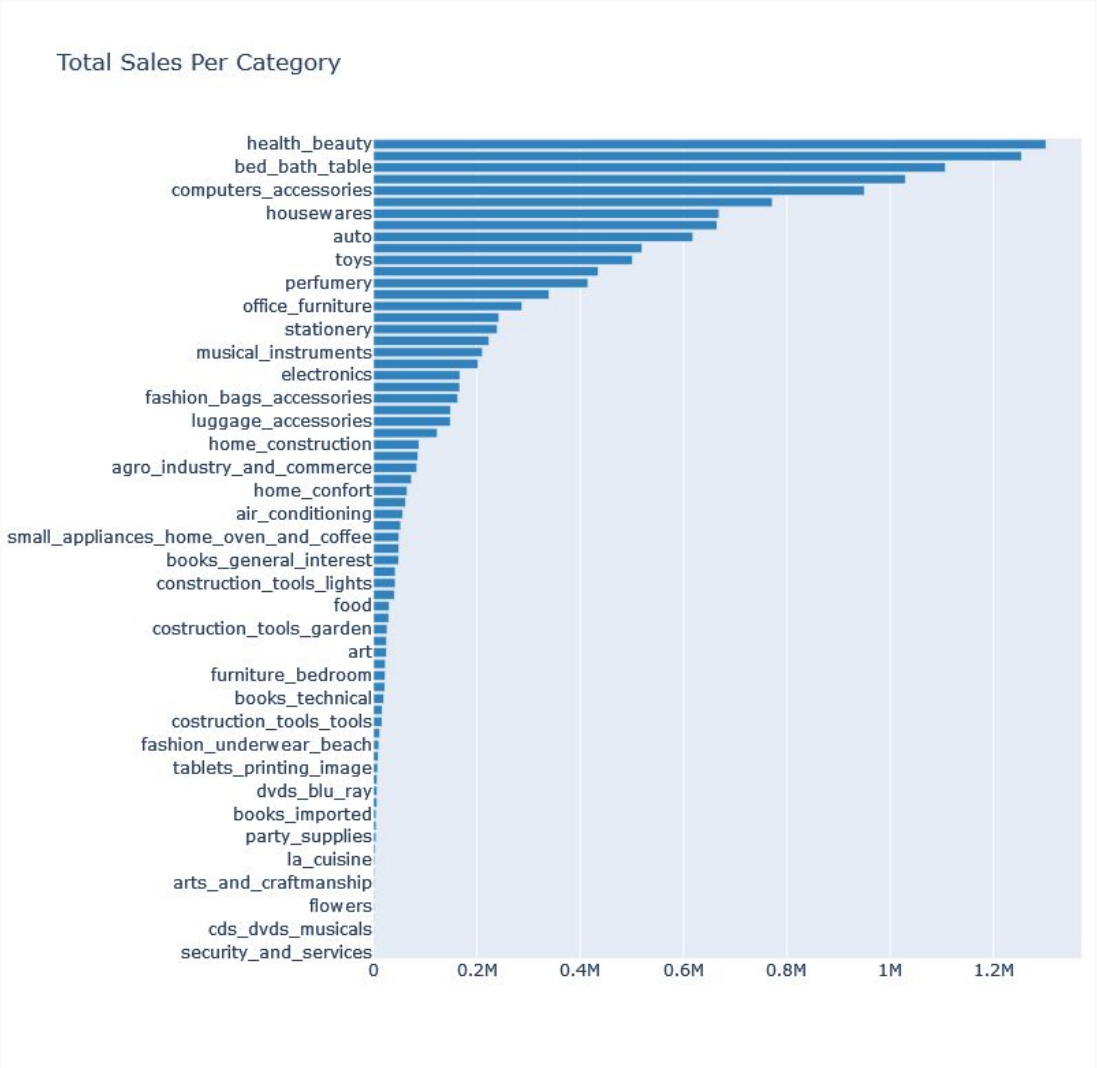
We only have very limited customers who purchase more than one items, but we have plenty of vendors selling multiple products.

This motivates us to measure category similarities based on the assumptions that **if two categories of products are sold by the same seller, they tends to be more similar.**

Category Ranking by Total Sales

Top 5 categories:

- Health & beauty
- Bed & bath table
- Computers accessories
- Housewares
- Auto



Examples of Category Groups

— Similar categories are grouped together as expected

Group 1



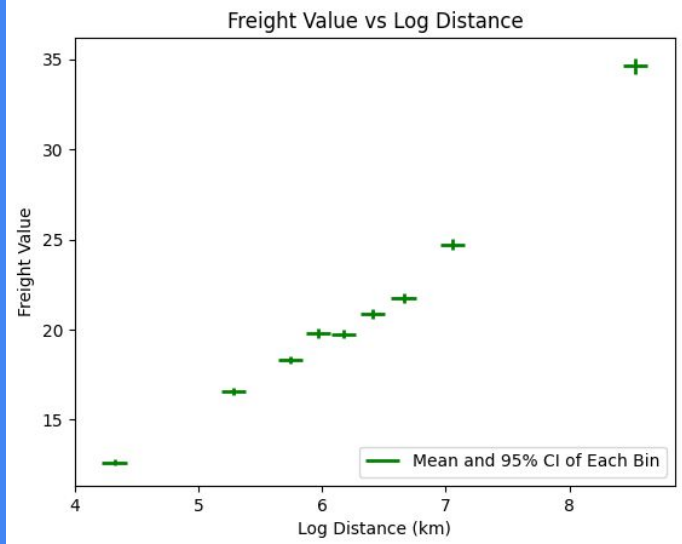
Group 2



Part 3: Intrinsic Product Features

The following product features are taken into consideration:

- Average ratings
- Price difference between products and customers' historical purchases
- Distance between product sellers and customers



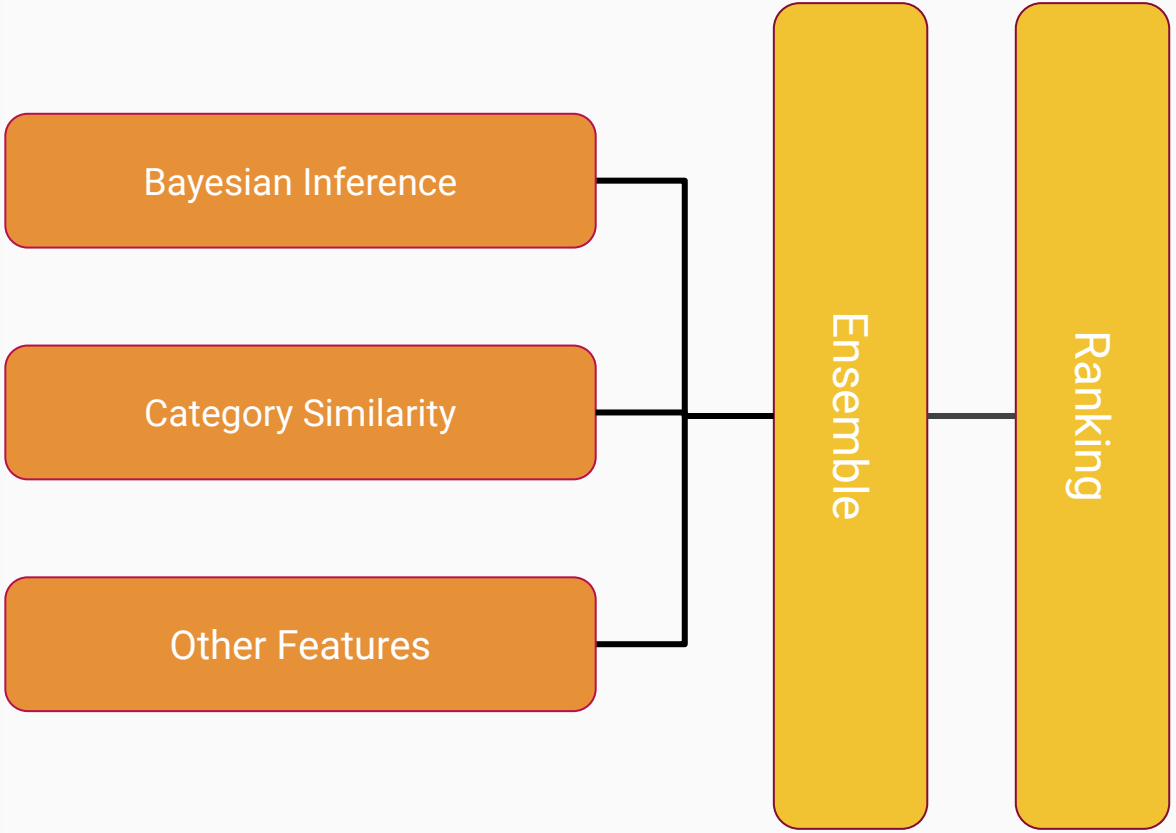
Validate the importance of selected features



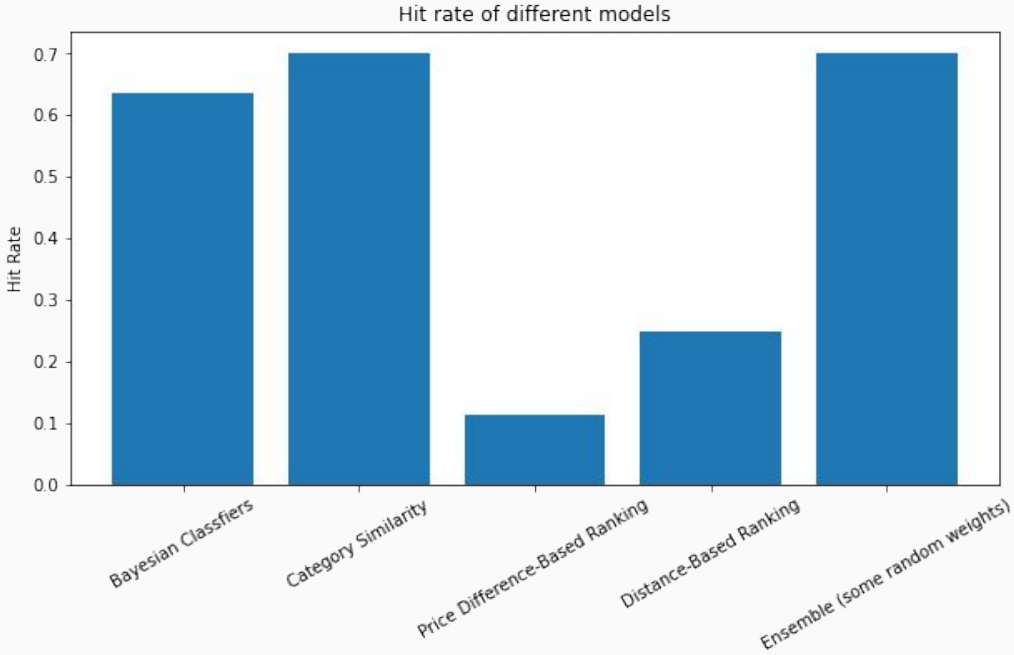
Final Part: Ensemble Model

Ensemble takes advantage of different models and exploits different data information.

The weights for different models are trained (e.g. grid search) using the limited customer purchase history.



Final Results



- All models have some classification power
- Category similarity has the best predicting power
- Since we have limited customer-product interactions, the performance is not yet satisfying for industrial requirements

Thank you!

Appendix: Time Series Analysis for Sales

Increased Sales in 2018



Appendix: Price Distribution by Payment Types

78% payment through credit cards, mean price is R$153

